

OMKAR KADAM

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OBJECTIVE

I recently completed my BE in Information Technology from Terna Engineering College (Mumbai University, 2026), and I work as a Full Stack Engineer at Riamona Luxury & Fashion Brands, where I build full-stack products end to end – writing the code, the tests, the CI/CD, and running them in production. Day to day I work with React, Node.js, Python, Prisma, and PostgreSQL. I recently placed 7th out of 31,000+ teams at the Meta PyTorch Hackathon and co-authored a research paper now under review at *Discover Internet of Things* (Springer Nature). I'm looking for a full-stack, platform, or ML engineering role.

EDUCATION

Terna Engineering College, Nerul – BE Information Technology, Mumbai University	Nov 2022 – June 2026
D.A.V. Public School, Airoli – CBSE 12th Science	June 2020 – July 2022
D.A.V. Public School, Airoli – CBSE 10th	May 2020

EXPERIENCE

Full Stack Engineer – Riamona Luxury and Fashion Brands Pvt. Ltd., *Navi Mumbai* Jan 2026 – Present

- Build and ship full-stack features for several internal products using React, Node.js, Python, and PostgreSQL, along with the REST/GraphQL APIs and database schemas behind them
- Write unit, integration, and end-to-end tests for what I build, and maintain the GitHub Actions and Railway pipelines that run them on every push
- Handle deployments, monitoring, and production issues across staging and production, and work with the team to turn requirements into shipped features

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, Java, SQL, HTML, CSS
AI Engineering	RAG, LLM APIs, MCP, LangChain, pgvector
ML & Data	PyTorch, TensorFlow, Scikit-learn, NLTK, OpenCV, Pandas, NumPy, Streamlit, Deep Learning, NLP, Computer Vision
Web & Frameworks	React, Next.js, Node.js, Express.js, NestJS, Flask, FastAPI, Prisma, TailwindCSS, Vite, Three.js, WebSockets
Databases	PostgreSQL, MySQL, MongoDB, Supabase, Redis, SQLAlchemy
Cloud & DevOps	Google Cloud, AWS, Docker, Railway, GitHub Actions, Git
APIs & Integration	REST APIs, GraphQL, JWT, OAuth, Shopify Admin API, BeautifulSoup
Blockchain	Solidity, Web3.js, Hardhat
Testing & Monitoring	pytest, Vitest, Playwright, Sentry

PUBLICATIONS

Vehicle-to-Vehicle Communication for Blind Spot Detection and Accident Prevention Using Severity-Gated Collision Risk Indexing – Co-author, manuscript submitted to *Discover Internet of Things* (Springer Nature); currently under peer review. April 2026

PROJECTS

- **VayuNetra – AI-Powered Urban Air Quality Intelligence (July 2026):** A six-agent AI platform that turns air quality readings into attributed, delivered interventions across Delhi, Bengaluru, and Mumbai. Does per-km² PM2.5 source attribution (GBM + SHAP), 72-hour calibrated forecasts, RAG-cited enforcement dossiers with draft notice PDFs, and citizen advisories in 4 languages over app, Telegram, and IVR calls – 6,394 modeled cells running on zero-cost infrastructure. [GitHub](#)

- **Vehicle-to-Vehicle Communication for Blind Spot Detection and Accident Prevention Using Severity-Gated Collision Risk Indexing (May 2026):** A simulation where vehicles warn each other about blind spots and likely collisions before they happen. Built the messaging between vehicles, the sensor data processing, and a risk score that decides when to alert the driver. The work became a research paper, now under review at *Discover Internet of Things* (Springer Nature). [GitHub](#)
- **Smart PUC (May 2026):** A blockchain-based system for monitoring vehicle emissions and PUC compliance in India. OBD devices sign live emission telemetry, testing stations validate it, and the records go on-chain so no single party can fake the data. Tracks all five Bharat Stage VI pollutants with physics-based models and an ML ensemble for fraud detection, and issues emission certificates as NFTs. Built with Solidity smart contracts (Hardhat, OpenZeppelin), a FastAPI/Web3.py backend, and a web dashboard. [GitHub](#)
- **Shopify Gold Price Automation System (March 2026):** End-to-end automated pricing engine for a luxury jewellery Shopify store with ~939 products and 55,000+ variants. Scrapes live gold rates from IBJA twice daily, recalculates all variant prices via a component-wise formula (gold weight $\times$ purity rate + diamond weight + making charges), and pushes updates through Shopify GraphQL Bulk Operations API with zero manual steps. Deployed on Railway with Flask, PostgreSQL, and Telegram bot notifications. (*Proprietary*)
- **Invoice & Payment Management Portal (March 2026):** Full-stack enterprise financial platform for invoice tracking, payment scheduling, and UTR verification. Features a slab-based early-payment discount engine, daily interest penalty calculations, bulk Excel invoice import, role-based access (Admin, Customer, Agent, Sub-Agent, Employee), and a payment simulator. Built with React, TypeScript, Express.js, Prisma ORM, and PostgreSQL; deployed on Railway. (*Proprietary*)
- **Face Recognition Attendance System (May 2025):** An attendance system that marks students present from a camera feed using face recognition. Built with OpenCV and the face-recognition library, with records stored in MySQL and exported as attendance reports. [GitHub](#)
- **Fitness Tracker Application (Nov 2024):** A web app for logging diet, sleep, and workouts, with a BMI calculator and goal tracking. Uses Chart.js to show progress over time. [GitHub](#)
- **Movie Recommendation System (May 2024):** A movie recommendation web app built on collaborative filtering. Pulls movie data from the IMDB/TMDB APIs and runs sentiment analysis on user reviews with NLTK. React frontend, Flask backend. [GitHub](#)

LEADERSHIP / EXTRA-CURRICULAR ACTIVITIES

- **7th Place – Meta PyTorch Hackathon (Scaler School of Technology):** Placed 7th out of 31,000+ teams. The hackathon ran in two rounds – an online round with 31,000+ teams, then an offline finals with the top 800. Built *Chakravyuh*, a multi-agent reinforcement learning environment for detecting UPI payment fraud, with LoRA-fine-tuned agents trained in PyTorch using GRPO. [GitHub](#)
- **Finalist – Avishkar Research Project Competition 2025, Mumbai University:** Reached the finals of Avishkar 2025, Mumbai University's inter-collegiate research competition, presenting the V2V research project against hundreds of teams.
- **Computer Society of India, Terna Chapter – Advisor (Oct 2025 – Present), Vice Chairperson (Oct 2024 – Oct 2025), Technical Executive (Nov 2023 – Oct 2024):** Started as Technical Executive and moved up to Vice Chairperson over 2.5 years. As Vice Chairperson, ran a 40-member team that organised 10+ AI/ML workshops and seminars. Now an Advisor, helping the new committee settle in.
- **Training & Placement Cell, Terna – Deputy Secretary (Nov 2024 – Oct 2025), Hospitality Committee Member (Nov 2023 – Nov 2024):** Joined as a Hospitality Committee Member and was promoted to Deputy Secretary. Worked with 20+ HR professionals and recruiters and organised mock interviews to get students placement-ready.
- **Revive Cultural Fest, Terna – Hospitality HOD (Jan 2025 – Jan 2026), Senior Member (Jan 2024 – Jan 2025), Committee Member (Dec 2022 – Jan 2024):** Worked my way up from Committee Member to Hospitality HOD over 3 years at Revive, the college's cultural fest. As HOD, ran hospitality for 1,000+ attendees and guest judges with a team of 50+ volunteers.